

CPT204 Mock Exam Questions 13-14 - Completed Suggested Answers

Question 13 - OnlineCourse

```
interface Payable {
    double getPaymentAmount();
}

class OnlineCourse implements Payable {
    private String courseName;
    private double baseFee;
    private Level level;
    private boolean hasCertificate;

    public enum Level {
        BEGINNER, INTERMEDIATE, ADVANCED
    }

    public OnlineCourse(String courseName, double baseFee,
                        Level level, boolean hasCertificate) {
        this.courseName = courseName;
        this.baseFee = baseFee;
        this.level = level;
        this.hasCertificate = hasCertificate;
    }

    @Override
    public double getPaymentAmount() {
        if (baseFee < 0) {
            return 0;
        }

        double amount = baseFee;
        if (level == Level.INTERMEDIATE) {
            amount += 100;
        } else if (level == Level.ADVANCED) {
            amount += 200;
        }

        if (hasCertificate) {
            amount += 50;
        }
        return amount;
    }
}
```

Question 14 - KeywordCounter

```
import java.util.*;

interface Countable {
    int getKeywordCount();
}

class KeywordCounter implements Countable {
    private String text;
    private HashSet<String> keywords;
    private HashMap<String, Integer> wordCounts;

    public KeywordCounter(String text, Collection<String> keywords) {
        this.text = text;
        this.keywords = new HashSet<>(keywords);
    }
}
```

```

        this.wordCounts = new HashMap<>();
        countWords();
    }

    private void countWords() {
        if (text == null || text.isEmpty()) {
            return;
        }

        String[] words = text.split(" ");
        for (String word : words) {
            if (wordCounts.containsKey(word)) {
                wordCounts.put(word, wordCounts.get(word) + 1);
            } else {
                wordCounts.put(word, 1);
            }
        }
    }

    @Override
    public int getKeywordCount() {
        if (text == null || text.isEmpty()) {
            return 0;
        }

        int total = 0;
        for (String keyword : keywords) {
            if (wordCounts.containsKey(keyword)) {
                total += wordCounts.get(keyword);
            }
        }
        return total;
    }
}

```